

2c. Content of non exam assessment Programming project (Component 03 or 04)

Learners will be expected to analyse, design, develop, test, evaluate and document a program written in a suitable programming language. The underlying approach to the project is to apply the principles of computational thinking to a practical coding problem. Learners are expected to apply appropriate principles from an agile development approach to the project development.

While the project assessment criteria are organised into specific categories, it is anticipated the final report will document the agile development process and elements for each of the assessment categories will appear throughout the report.

3.1. Analysis of the problem (10 marks)	
3.1.1 Problem identification	(a) Describe and justify the features that make the problem solvable by computational methods. (b) Explain why the problem is amenable to a computational approach.
3.1.2 Stakeholders	(a) Identify and describe those who will have an interest in the solution explaining how the solution is appropriate to their needs (this may be named individuals, groups or persona that describes the target end user).
3.1.3 Research the problem	(a) Research the problem and solutions to similar problems to identify and justify suitable approaches to a solution. (b) Describe the essential features of a computational solution explaining these choices. (c) Explain the limitations of the proposed solution.
3.1.4 Specify the proposed solution	(a) Specify and justify the solution requirements including hardware and software configuration (if appropriate). (b) Identify and justify measurable success criteria for the proposed solution.

3.2 Design of the solution (15 marks)	
3.2.1 Decompose the problem	(a) Break down the problem into smaller parts suitable for computational solutions justifying any decisions made.
3.2.2 Describe the solution	(a) Explain and justify the structure of the solution. (b) Describe the parts of the solution using algorithms justifying how these algorithms form a complete solution to the problem. (c) Describe usability features to be included in the solution. (d) Identify key variables / data structures / classes justifying choices and any necessary validation.
3.2.3 Describe the approach to testing	(a) Identify the test data to be used during the iterative development and post development phases and justify the choice of this test data.
